Special Summer Integrated Learning Programme

Mathematics

Polynomial Functions and Rational Functions

Week 2 Day 1 AM

SINGAPORE UNIVERSITY OF TECHNOLOGY AND DESIGN

Outline

1. Quadratic Functions
2. Polynomials, The Remainder and Factor Theorems
3. The Intermediate Value Theorem for Polynomials and The Rational Zeros Theorem
4. Summary

Quadratic functions

Definition:

A ***quadratic function*** is a polynomial function of degree 2. So a quadratic function is a function of the form

$$f(x) = ax^2 + bx + c, \quad \text{where } a \neq 0.$$

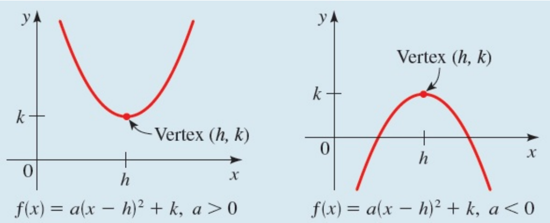
By completing the square, we can always rewrite a quadratic function in the ***standard form***:

$$\begin{aligned} f(x) = ax^2 + bx + c &= a\left(x^2 + \frac{b}{a}x\right) + c \\ &= a\left(x^2 + \frac{b}{a}x + \left(\frac{b}{2a}\right)^2\right) - a\cdot\left(\frac{b}{2a}\right)^2 + c \\ &= a\left(x + \frac{b}{2a}\right)^2 + c - \frac{b^2}{4a} = a(x-h)^2 + k, \end{aligned}$$

where $h = -\frac{b}{2a}$ and $k = c - \frac{b^2}{4a}$.

Graphing quadratic functions

Since the standard form $f(x) = a(x-h)^2 + k$ is just three transformations (horizontal shifting by h, vertical stretching by a factor of a, and vertical shifting by k) from $y = x^2$, we can easily graph the standard form of a quadratic function:

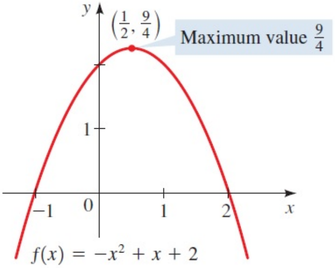


Remarks:

- The graph is symmetric w.r.t. the vertical line $x = h$, so h is the mid-value of the two roots of the quadratic function.
- When $a > 0$ (resp. $a < 0$), the quadratic function has a global minimum (resp. maximum) at (h, k), but no global maximum (resp. minimum).

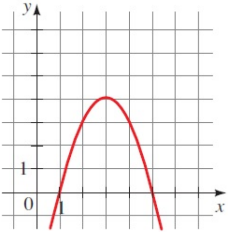
Graphing quadratic functions - example

E.g.: To graph $f(x) = -x^2 + x + 2$, we first complete the square and get $f(x) = -\left(x - \frac{1}{2}\right)^2 + \frac{9}{4}$. Thus, the graph has an $\cap$ shape with a global maximum at $\left(\frac{1}{2}, \frac{9}{4}\right)$. To find the x-intercepts, we factorize the quadratic function:
$f(x) = -x^2 + x + 2 = -(x^2 - x - 2) = -(x+1)(x-2)$, so the x-intercepts are -1 and 2.



Activity 1

(a) Given the graph of a quadratic function $f(x)$ (drawn to scale with a global maximum at $(3, 4)$):



find the expression of $f(x)$.

(b) Find the maximum or minimum value of $g(t) = -3 + 80t - 20t^2$, for $t \in \mathbb{R}$.

(c) Redo the previous part, but we change the domain to $[3, 5]$.

Activity 1: solution

(a) With the global maximum $(3,4)$, we conclude that $f(x)=a(x-3)^2+4$. By using the x-intercept 1, we have $0=a(1-3)^2+4 \;\Leftrightarrow\; -4=4a \;\Leftrightarrow\; a=-1$. Hence, $f(x)=-(x-3)^2+4$.
Alternatively, since the x-intercepts are 1 and 5, we conclude that $f(x)=a(x-1)(x-5)$. Using the coordinates of the global maximum $(3,4)$, we have
$4=a(3-1)(3-5) \;\Leftrightarrow\; 4=-4a$, which gives $a=-1$.
Hence, $f(x)=-(x-1)(x-5)=-x^2+6x-5$.

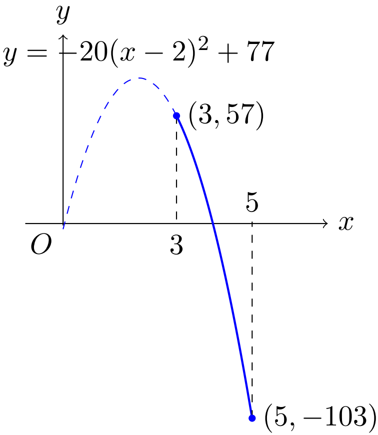
(b) We complete the square and get

$$\begin{aligned} g(t) &= -3+80t-20t^2 = -20(t^2-4t)-3 \\ &= -20\left(t^2-4t+4-4\right)-3 \\ &= -20(t-2)^2+80-3 = -20(t-2)^2+77. \end{aligned}$$

Thus, $g(x)$ has a global maximum at $(2,77)$, i.e. the maximum value of g is 77.

Activity 1: solution

(c) Since $g(t) = -20(t-2)^2 + 77$, we can graph the quadratic function and then look at the part over the interval $3 \le x \le 5$:



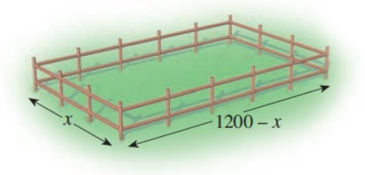
Hence, with the domain $= [3, 5]$, the quadratic function has the maximum value $= 57$ and the minimum value $= -103$.

Activity 2

(a) You have 2400 ft of fencing to enclose a rectangular horse pen.

  (i) Find a function $f(x)$ that models the area of the pen in terms of the width x of the pen (see the picture below). Remember to state the domain of f.

  (ii) Find the dimensions of the pen that maximize the area of the pen and give the maximum area also.



(b) Now suppose the pen must have a 100-ft-wide gate on one of its sides, so that this portion does not require fencing. Determine the dimensions of the pen that maximize its area, and find the corresponding maximum area.

Activity 2: solution

(a) (i) $f(x) = (1200 - x)x$ (area of a rectangle is the product of the length and width). Since distance ≥ 0, we must have $0 \leq x \leq 1200$. So the domain of f is $[0, 1200]$.

(ii) We complete the square:

$$\begin{aligned} f(x) &= -x^2 + 1200x \\ &= -(x^2 - 1200x) \\ &= -(x^2 - 1200x + 600^2 - 600^2) \\ &= -(x - 600)^2 + 600^2. \end{aligned}$$

We check that the global maximizer $x = 600$ is indeed in the domain $[0, 1200]$. Hence, the maximum area of the pen is $600^2 = 360000$sq ft, which is attained when the width is 600 ft and the length is $1200 - 600 = 600$ ft.

Activity 2: solution

(b) If we let the length and width of the pen be y and x, then

$$2x + 2y - 100 = 2400 \quad \Leftrightarrow \quad y = 1250 - x.$$

So the area of the pen is $A(x) = (1250 - x)x$ with domain $[0, 1250]$. By completing the square, we get

$$\begin{aligned} A(x) &= -x^2 + 1250x \\ &= -(x^2 - 1250x) \\ &= -(x^2 - 1200x + 625^2 - 625^2) \\ &= -(x - 625)^2 + 625^2. \end{aligned}$$

Hence, the maximum area of the pen with a 100-ft-wide gate is $625^2 = 390625$sq ft, which is attained when the width $x = 625$ ft and the length $y = 1250 - 625 = 625$ ft.

Outline

Polynomial functions

Definition:

A ***polynomial function of degree*** n is a function of the form

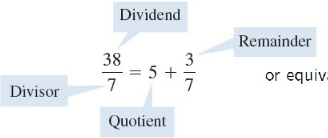
$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0,$$

where n is a natural number and an $a_n \neq 0$. The number a_0 is called the ***constant coefficient*** or ***constant term***. The number a_n is called the ***leading coefficient***.

Fun fact: In modern computer design programs, such as Adobe Illustrator or Microsoft Paint, curves are modelled by piecing together parts of cubic polynomials. A curve can be drawn by fixing two points, then using the mouse to drag one or more anchor points. Moving the anchor points amounts to adjusting the coefficients of a cubic polynomial.

Division theorem of polynomials

Dividing polynomials is similar to the process of dividing numbers. When we divide 38 by 7, the quotient is 5 and the remainder is 3. We write



or equivalently $38 = 5 \cdot 7 + 3$

Division Theorem and Definition:

If $P(x)$ and $D(x)$ are polynomials, with $D(x) \neq 0$, then there exist unique polynomials $Q(x)$ and $R(x)$, where $R(x)$ is either 0 or of degree strictly less than the degree of $D(x)$, such that

$$\frac{P(x)}{D(x)} = Q(x) + \frac{R(x)}{D(x)} \quad \text{or equivalently} \quad P(x) = Q(x)D(x) + R(x).$$

$P(x)$ and $D(x)$ are called the ***dividend*** and ***divisor***, respectively, $Q(x)$ is the ***quotient***, and $R(x)$ is the ***remainder***.

Long division of polynomials

E.g.: If we divide $6x^2 - 26x + 12$ by $x - 4$, we get

$$\frac{\overbrace{6x^2 - 26x + 12}^{\text{Dividend}}}{\underbrace{x - 4}_{\text{Divisor}}} = \overbrace{6x - 2}^{\text{Quotient}} + \frac{\overbrace{4}^{\text{Remainder}}}{x - 4}$$

or

$$\underbrace{6x^2 - 26x + 12}_{\text{Dividend}} = \underbrace{(x - 4)}_{\text{Divisor}}\underbrace{(6x - 2)}_{\text{Quotient}} + \overbrace{4}^{\text{Remainder}}$$

But how do we get the quotient $= 6x - 2$ and the remainder $= 4$? We can find them by the long division of polynomials:

- We begin by arranging them as follows (similar to the long division of numbers):

  $$x - 4\overline{)6x^2 - 26x + 12}$$

  Make sure that you arrange the terms in descending power from left to right.

Long division of polynomials

- Next, we create the same leading term $6x^2$ as the dividend by multiplying $6x$ to the divisor $x-4$, which gives us $6x^2-24x$.
- After that, we subtract $6x^2-26x+12$ by $6x^2-24x$ and get $-2x+12$.

$$
\begin{array}{r l}
 & 6x \\
x-4 & \overline{)\,6x^2-26x+12} \\
 & \underline{6x^2-24x} \\
 & -2x+12
\end{array}
$$

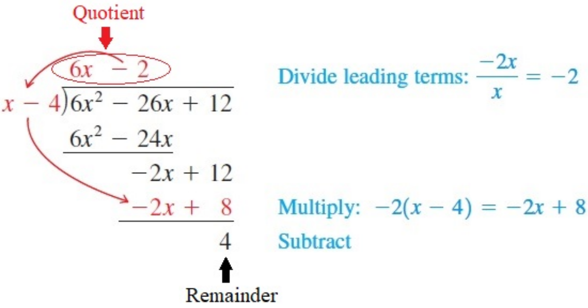
Divide leading terms: $\dfrac{6x^2}{x} = 6x$

Multiply: $6x(x-4) = 6x^2 - 24x$

Subtract and "bring down" 12

Long division of polynomials

- We repeat the process again, by creating the same leading term $-2x$ as $-2x+12$ by multiplying -2 to the divisor $x-4$, which gives us $-2x+8$. We then subtract $-2x+12$ by $-2x+8$ and get 4. Since 4 is a polynomial of degree $0 < 1 =$ degree of the divisor $x-4$, it's the remainder and we are done.



Activity 3

Find the quotient and remainder using long division.

(a) $\dfrac{x^3 + 2x^2 - x + 1}{x + 3}$

(b) $\dfrac{4x^3 + 2x^2 + 2x + 1}{2x + 1}$

(c) $\dfrac{x^3 + 2x + 1}{x^2 - x + 3}$

(d) $\dfrac{6x^3 + 2x^2 + 22x}{2x^2 + 5}$

Activity 3: solution

(a)

$$
\begin{array}{r|l}
 & x^2 - x + 2 \\
\hline
x+3 & x^3 + 2x^2 - x + 1 \\
 & x^3 + 3x^2 \\
\hline
 & -x^2 - x \\
 & -x^2 - 3x \\
\hline
 & 2x + 1 \\
 & 2x + 6 \\
\hline
 & -5
\end{array}
$$

So

$$\frac{x^3 + 2x^2 - x + 1}{x + 3} = x^2 - x + 2 - \frac{5}{x + 3}.$$

Activity 3: solution

(b)

$$
\begin{array}{r|l}
 & \phantom{4x^3 + {}} 2x^2 + 0x + 1 \\
\cline{2-2}
2x+1 & 4x^3 + 2x^2 + 2x + 1 \\
 & \underline{4x^3 + 2x^2} \\
 & \phantom{4x^3 + 2x^2 + {}} 2x + 1 \\
 & \phantom{4x^3 + 2x^2 + {}} \underline{2x} + 1 \\
 & \phantom{4x^3 + 2x^2 + 2x + {}} 0
\end{array}
$$

So

$$\frac{4x^3 + 2x^2 + 2x + 1}{2x + 1} = 2x^2 + 1.$$

Activity 3: solution

(c)

$$
\begin{array}{r}
x + 1 \\
x^2 - x + 3 \,\big)\overline{\, x^3 + 0x^2 + 2x + 1} \\
\underline{x^3 - x^2 + 3x\phantom{{}+1}} \\
x^2 - x + 1 \\
\underline{x^2 - x + 3} \\
-2
\end{array}
$$

So

$$\frac{x^3 + 2x + 1}{x^2 - x + 3} = x + 1 - \frac{2}{x^2 - x + 3}.$$

Activity 3: solution

(d)

$$\begin{array}{r|r}
 & 3x + 1 \\ \hline
2x^2 + 0x + 5 & 6x^3 + 2x^2 + 22x + 0 \\
 & \underline{6x^3 + 0x^2 + 15x} \phantom{{}+0} \\
 & 2x^2 + 7x + 0 \\
 & \underline{2x^2 + 0x + 5} \\
 & 7x - 5
\end{array}$$

So

$$\frac{6x^3 + 2x^2 + 22x}{2x^2 + 5} = 3x + 1 + \frac{7x - 5}{2x^2 + 5}.$$

The Remainder and Factor Theorems

The division theorem of polynomials implies the two following important theorems.

The Remainder Theorem

$P(c)$ is the remainder of $P(x)$ divided by $x - c$.

Proof: The remainder is a number because its degree $< 1 =$ degree of the divisor $x - c$, so let the remainder be r and we have $P(x) = Q(x)(x - c) + r$. Substitute $x = c$ and we get $P(c) = Q(c) \cdot 0 + r \iff P(c) = r$.

The Factor Theorem

$P(c) = 0$ iff. $x - c$ is a factor of $P(x)$.

Proof: This follows immediately from the remainder theorem. Since $P(x) = Q(x)(x - c) + P(c)$, so $P(x) = Q(x)(x - c)$ iff. $P(c) = 0$.

The Remainder and Factor Theorems - examples

E.g.: For the example $\frac{P(x)}{x-4} = \frac{6x^2-26x+12}{x-4}$ on slide 15, we verify that $P(4) = 6 \cdot 4^2 - 26 \cdot 4 + 12 = 4$, which is indeed the remainder.

E.g.: Let $P(x) = 3x^4 - 2x^3 + 2x^2 + 2x - 5$. We check that $P(1) = 3 - 2 + 2 + 2 - 5 = 0$, hence the factor theorem says that $P(x) = (x-1)Q(x)$. By the long division, we get

$$
\begin{array}{r|l}
 & 3x^3 + x^2 + 3x + 5 \\
\hline
x - 1 & 3x^4 - 2x^3 + 2x^2 + 2x - 5 \\
 & 3x^4 - 3x^3 \\
\hline
 & x^3 + 2x^2 \\
 & x^3 - x^2 \\
\hline
 & 3x^2 + 2x \\
 & 3x^2 - 3x \\
\hline
 & 5x - 5 \\
 & 5x - 5 \\
\hline
 & 0
\end{array}
$$

So we have $3x^4 - 2x^3 + 2x^2 + 2x - 5 = (3x^3 + x^2 + 3x + 5)(x - 1)$.

Outline

Finding roots/zeros of polynomials

The last example shows that we can apply the factor theorem to help us to factorize polynomials, provided we can find c such that $P(c) = 0$. Recall that for a general function $f(x)$, a value c such that $f(c) = 0$ is called a ***root*** or a ***zero*** of f. So in order to apply the factor theorem to factorize a polynomial, we need to find a root of the polynomial.

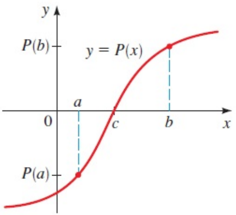
We can try to look for a root c by comparing the coefficients. For instance, is the sum of the coefficients equal to 0? If it is then $c = 1$ will work. Is the ***alternating sum*** (the sum in which the signs of the terms alternate between positive and negative) of the coefficients equal to 0? If it is then $c = -1$ will work.

The next theorem also helps to pin down a root within an interval (a, b).

The Intermediate Value Theorem (IVT) for polynomials

The Intermediate Value Theorem (IVT) for Polynomials

If $P(x)$ is a polynomial function and $P(a)$ and $P(b)$ have opposite signs, then there exists at least one value c between a and b for which $P(c) = 0$.



E.g.: Consider $P(x) = 3x^5 - 5x^4 + 4x^3 - 7x + 3$. Since $P(0) = 3 > 0$ and $P(1) = 3 - 5 + 4 - 7 + 3 = -2 < 0$, so there must be a root c in the interval $(0, 1)$.

Activity 4

Factorize $P(x) = 3x^4 - 2x^3 + 2x^2 + 2x - 5$ completely from slide 24.

Hint: since from slide 24, we've shown

$$P(x) = (3x^3 + x^2 + 3x + 5)(x - 1),$$

so we just need to factorize $Q(x) := 3x^3 + x^2 + 3x + 5$.

Activity 4: solution

We observe that the alternating sum of the coefficients $3-1+3-5=0$, so we try $c=-1$: $Q(-1)=-3+1-3+5=0$. Hence, $x+1$ is a factor of $Q(x)$. By the long division, we get

$$
\begin{array}{r|l}
 & 3x^2 - 2x + 5 \\
\hline
x+1 & 3x^3 + x^2 + 3x + 5 \\
 & \underline{3x^3 + 3x^2} \\
 & -2x^2 + 3x \\
 & \underline{-2x^2 - 2x} \\
 & 5x + 5 \\
 & \underline{5x + 5} \\
 & 0
\end{array}
$$

So

$$Q(x)=(3x^2-2x+5)(x+1).$$

Activity 4: solution

Next, we need to factorize $R(x) := 3x^2 - 2x + 5$. But the discriminant $D = (-2)^2 - 4 \cdot 3 \cdot 5 = -56 < 0$, so $R(x)$ has no real roots and can't be factorized (with real coefficients).

So the complete factorization of $P(x)$ is

$$3x^4 - 2x^3 + 2x^2 + 2x - 5 = (x+1)(x-1)(3x^2 - 2x + 5).$$

The Rational Zeros Theorem

After applying the IVT to narrow down a root of a polynomial within an interval, what numbers in the interval should we try? In general, you can subdivide the interval and apply the IVT repeatedly, until the interval becomes small enough for the accuracy that you want. This numerical method of finding roots of functions is called the bisection method.

If $P(x)$ is a polynomial with integer coefficients and it has a rational root/zero, then we can apply the following theorem to find it (same idea as the quadratic factorization that we saw last week).

The Rational Zeros Theorem

If the polynomial $P(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$ has integer coefficients (where $a_n \neq 0$ and $a_0 \neq 0$), then every rational root/zero of P is of the form $\frac{p}{q}$, where p, q are integers such that p is a factor of a_0 and q is a factor of a_n.

Please refer to page 276 of the textbook for the proof.

The Rational Zeros Theorem - example

E.g.: Suppose we want to factorize $P(x) = 3x^4 - 13x^3 - 7x^2 - 13x - 10$. We quickly check that $P(-1) = 3 + 13 - 7 + 13 - 10 = 12 > 0$ and $P(0) = -10 < 0$, so there must be at least a root in the interval $(-1, 0)$. If the root is rational, then by the rational zeros theorem, it must be

$$\frac{p}{q},$$

where p is a factor of -10 and q is a factor of 3. The only choices that fall in the interval $(-1, 0)$ are $-\frac{2}{3}$ and $-\frac{1}{3}$ (the others are $\pm\frac{10}{3}, \pm\frac{5}{3}, \frac{2}{3}, \frac{1}{3}, \pm 10, \pm 5, \pm 2, \pm 1$).
Since $P\left(-\frac{2}{3}\right) = 0$ and $P\left(-\frac{1}{3}\right) = -\frac{160}{27}$, $-\frac{2}{3}$ is a root of $P(x)$, i.e. $x - \left(-\frac{2}{3}\right) = x + \frac{2}{3}$ is a factor of $P(x)$. Since multiplying $x + \frac{2}{3}$ by any non-zero constant is still a factor of $P(x)$, we will divide $P(x)$ by $3x + 2$ using the long division.

The Rational Zeros Theorem - example

$$
\begin{array}{r|l}
 & x^3 - 5x^2 + x - 5 \\
\hline
3x+2 & 3x^4 - 13x^3 - 7x^2 - 13x - 10 \\
 & \underline{3x^4 + 2x^3} \\
 & -15x^3 - 7x^2 \\
 & -\underline{15x^3 - 10x^2} \\
 & 3x^2 - 13x \\
 & \underline{3x^2 + 2x} \\
 & -15x - 10 \\
 & -\underline{15x - 10} \\
 & 0
\end{array}
$$

So

$$P(x) = 3x^4 - 13x^3 - 7x^2 - 13x - 10 = (3x+2)(x^3 - 5x^2 + x - 5).$$

ACTIVITY 5

Find all the rational roots of the following polynomials.

(a) $P(x) = 3x^4 - 13x^3 - 7x^2 - 13x - 10$ (the polynomial from the previous example.)

(b) $Q(x) = 3x^4 - 10x^3 - 9x^2 + 40x - 12$

(c) $R(x) = 3x^4 + 4x^3 - 7x^2 - 2x - 3$

Activity 5: solution

(a) Since
$P(x) = 3x^4 - 13x^3 - 7x^2 - 13x - 10 = (3x+2)(x^3 - 5x^2 + x - 5)$, so we proceed to factorize $P_1(x) := x^3 - 5x^2 + x - 5$. By the rational zeros theorem, the potential rational roots are $\pm 1, \pm 5$. Since $P_1(-5) = -260, P_1(-1) = -12, P_1(1) = -8$ and $P_1(5) = 0$, so $x - 5$ is a factor of $P_1(x)$. By the long division, we have

$$\begin{array}{r|l} & x^2 + 0x + 1 \\ \hline x - 5 & x^3 - 5x^2 + x - 5 \\ & x^3 - 5x^2 \\ \hline & x - 5 \\ & x - 5 \\ \hline & 0 \end{array}$$

So

$$P(x) = 3x^4 - 13x^3 - 7x^2 - 13x - 10 = (3x+2)(x-5)(x^2+1).$$

Since $x^2 + 1$ has no real roots, we conclude that $-\frac{2}{3}$ and 5 are all the rational roots of $P(x)$.

Activity 5: solution

(b) $Q(x) = 3x^4 - 10x^3 - 9x^2 + 40x - 12$. Since $Q(0) = -12$ and $Q(1) = 3 - 10 - 9 + 40 - 12 = 12$, there must be at least a root in $(0, 1)$. By the rational zeros theorem, the potential rational roots in this interval are $\frac{1}{3}$ and $\frac{2}{3}$. Since $Q\left(\frac{1}{3}\right) = 0$ and $Q\left(\frac{2}{3}\right) = \frac{224}{27} \neq 0$, so $3x - 1$ is a factor of $Q(x)$, but not $3x - 2$. By the long division, we have

$$
\begin{array}{r|l}
 & x^3 - 3x^2 - 4x + 12 \\
\hline
3x - 1 & 3x^4 - 10x^3 - 9x^2 + 40x - 12 \\
 & 3x^4 - x^3 \\
\hline
 & -9x^3 - 9x^2 \\
 & -9x^3 + 3x^2 \\
\hline
 & -12x^2 + 40x \\
 & -12x^2 + 4x \\
\hline
 & 36x - 12 \\
 & 36x - 12 \\
\hline
 & 0
\end{array}
$$

So
$Q(x) = 3x^4 - 10x^3 - 9x^2 + 40x - 12 = (3x - 1)(x^3 - 3x^2 - 4x + 12)$.

Activity 5: solution

(b) $Q(x) = (3x-1)(x^3-3x^2-4x+12)$. Let $Q_1(x) = x^3-3x^2-4x+12$. By the rational zeros theorem, the potential rational roots of $Q_1(x)$ are $\pm 12, \pm 6, \pm 4, \pm 3, \pm 2, \pm 1$. After evaluating $Q_1(x)$ at these values, we find $Q_1(-2) = Q_1(2) = Q_1(3) = 0$. So $(x+2)(x-2)(x-3)$ is a factor of $Q_1(x)$. But since both of them have the same degree and leading coefficient, they must be equal. Thus,

$$Q(x) = 3x^4-10x^3-9x^2+40x-12 = (3x-1)(x+2)(x-2)(x-3),$$

we conclude that $-2, \frac{1}{3}, 2, 3$ are all the rational roots of $Q(x)$.

Activity 5: solution

(c) $R(x) = 3x^4 + 4x^3 - 7x^2 - 2x - 3$. By the rational zeros theorem, the potential rational roots are $\pm 1, \pm 3, \pm \frac{1}{3}$. However, we verify that $R(x) \neq 0$ at these values. Hence, we conclude that $R(x)$ has no rational roots.

⚠ This doesn't mean that $R(x)$ has no real roots. In fact, $R(-3) = 75$, $R(-2) = -11$ and $R(2) = 45$, so there must be roots in the intervals $(-3, -2)$ and $(-2, 2)$.

Outline

Summary

- Quadratic Functions
  - quadratic functions and graphing them.
- Polynomials, The Remainder and Factor Theorems
  - polynomial functions.
  - division theorem of polynomials.
  - long division of polynomials.
  - the remainder and factor theorems.
- The Intermediate Value Theorem for Polynomials and The Rational Zeros Theorem
  - finding roots/zeros of polynomials.
  - the intermediate value theorem (IVT) for polynomials.
  - the rational zeros theorem.

Resources

References:

- James Stewart. Precalculus Mathematics for Calculus (7th edition) sections 3.1, 3.3, 3.4. Section 3.5 is not covered in this course, but it's good to know.